



32

STRATHMORE INSTITUTE OF MATHEMATICAL SCIENCES
MASTER OF SCIENCE IN DATA SCIENCE & ANALYTICS
CAT 1- Open Book
DSA 8505: Bayesian Statistics
SOLUTIONS

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1. A researcher is studying the probability θ that a patient responds to a new treatment. Before collecting data, the researcher specifies the prior distribution:

$$\theta \sim \text{Beta}(\alpha, \beta), \quad \alpha = 3, \beta = 7.$$

A clinical trial is conducted on $n = 20$ patients and $k = 14$ respond successfully.

Required:

- a. Write the likelihood function $p(D | \theta)$ under the Binomial model.

(2 MKS)

Solution

(a) Likelihood function

The likelihood function describes the probability of observing the data D given a particular value of the parameter θ .

Step 1: Specify the sampling model

Each patient either responds successfully or does not respond to the treatment. Assuming:

- each patient represents an independent trial, and
- the probability of success is constant and equal to θ ,

the total number of successful responses follows a Binomial distribution:

$$K | \theta \sim \text{Binomial}(n, \theta), \quad n = 20.$$

Step 2: Write the Binomial probability mass function

For a Binomial random variable K , the probability of observing k successes out of n trials is

$$P(K = k | \theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}.$$

Substituting the observed values $k = 14$ and $n = 20$:

$$p(D | \theta) = \binom{20}{14} \theta^{14} (1 - \theta)^{20-14} = \binom{20}{14} \theta^{14} (1 - \theta)^6.$$

Step 3: Interpret the likelihood function

The likelihood function is viewed as a function of θ for the fixed observed data D . It measures how plausible different values of θ are in light of the observed outcome of 14 successful responses.

Step 4: Express the likelihood up to proportionality

The binomial coefficient $\binom{20}{14}$ does not depend on θ . Since Bayesian updating only requires the likelihood up to a proportionality constant, we write

$$p(D | \theta) \propto \theta^{14} (1 - \theta)^6.$$

Conclusion

Thus, the likelihood function under the Binomial model is

$$p(D | \theta) \propto \theta^{14} (1 - \theta)^6.$$

This form will be used directly in the application of Bayes' theorem to obtain the posterior distribution of θ .

- b. Using Bayes' theorem, derive the posterior distribution $p(\theta | D)$ showing all steps.

Solution

(b) Posterior derivation using Bayes' theorem

Bayes' theorem expresses the posterior density of θ given the data D as

$$p(\theta | D) = \frac{p(D | \theta) p(\theta)}{p(D)},$$

where

$$p(D) = \int_0^1 p(D | \theta) p(\theta) d\theta$$

is the marginal likelihood (normalizing constant).

Step 1: Write down the likelihood function

Given that $k = 14$ successes are observed out of $n = 20$ trials, the Binomial likelihood is

$$p(D | \theta) = \binom{20}{14} \theta^{14} (1 - \theta)^{20-14} = \binom{20}{14} \theta^{14} (1 - \theta)^6.$$

Step 2: Write down the prior density

The prior distribution is $\theta \sim \text{Beta}(3, 7)$. The Beta density is

$$p(\theta) = \frac{1}{B(3, 7)} \theta^{3-1} (1 - \theta)^{7-1} = \frac{1}{B(3, 7)} \theta^2 (1 - \theta)^6,$$

where $B(\alpha, \beta)$ is the Beta function.

Step 3: Apply Bayes' theorem

Substituting the likelihood and prior into Bayes' theorem:

$$p(\theta | D) = \frac{\binom{20}{14} \theta^{14} (1 - \theta)^6 \cdot \frac{1}{B(3,7)} \theta^2 (1 - \theta)^6}{p(D)}.$$

Step 4: Remove constants independent of θ

The terms $\binom{20}{14}$, $B(3,7)$, and $p(D)$ do not depend on θ , so they can be absorbed into a proportionality constant:

$$p(\theta | D) \propto \theta^{14} (1 - \theta)^6 \cdot \theta^2 (1 - \theta)^6.$$

Step 5: Combine powers of θ and $(1 - \theta)$

Using exponent rules:

$$\theta^{14} \cdot \theta^2 = \theta^{16}, \quad (1 - \theta)^6 \cdot (1 - \theta)^6 = (1 - \theta)^{12}.$$

Therefore,

$$p(\theta | D) \propto \theta^{16} (1 - \theta)^{12}.$$

- c. State the posterior distribution clearly.

Solution

(c) Posterior distribution

We now state the posterior distribution of θ after observing the data D .

Step 1: Recall the posterior kernel

From part (b), the posterior density of θ (up to a normalizing constant) is

$$p(\theta | D) \propto \theta^{16} (1 - \theta)^{12}, \quad 0 < \theta < 1.$$

Step 2: Recall the general form of the Beta distribution

The probability density function of a Beta distribution with parameters α^* and β^* is

$$\text{Beta}(\alpha^*, \beta^*) : p(\theta) = \frac{1}{B(\alpha^*, \beta^*)} \theta^{\alpha^* - 1} (1 - \theta)^{\beta^* - 1}, \quad 0 < \theta < 1.$$

Step 3: Match the posterior kernel to the Beta form

Comparing

$$\theta^{\alpha^* - 1} (1 - \theta)^{\beta^* - 1} \quad \text{with} \quad \theta^{16} (1 - \theta)^{12},$$

we obtain

$$\alpha^* - 1 = 16 \Rightarrow \alpha^* = 17, \quad \beta^* - 1 = 12 \Rightarrow \beta^* = 13.$$

Step 4: State the posterior distribution

Therefore, the posterior distribution of θ given the observed data D is

$$\theta | D \sim \text{Beta}(17, 13).$$

Interpretation

The posterior parameters (17, 13) are obtained by adding the observed number of successes (14) and failures (6) to the prior parameters (3, 7), respectively. This reflects Bayesian learning: the prior belief is updated by the information contained in the data.

2. In a hospital, the number of emergency cases arriving per night is assumed:

$$Y \mid \lambda \sim \text{Poisson}(\lambda).$$

Assume a Gamma prior:

$$\lambda \sim \Gamma(\alpha, \beta), \quad \alpha = 4, \beta = 2.$$

In one night, $y = 7$ emergency cases are observed.

Required:

a. Derive the posterior distribution for λ .

(7 marks)

Solution

(a) Posterior distribution

We derive the posterior distribution of λ using Bayes' theorem.

Step 1: State Bayes' theorem

Bayes' theorem for the parameter λ given the observation y is

$$p(\lambda \mid y) = \frac{p(y \mid \lambda) p(\lambda)}{p(y)},$$

✓ ①

where

$$p(y) = \int_0^\infty p(y \mid \lambda) p(\lambda) d\lambda$$

is the marginal likelihood (normalizing constant).

Step 2: Write down the likelihood function

The conditional distribution of Y given λ is Poisson:

$$Y \mid \lambda \sim \text{Poisson}(\lambda).$$

Thus, the likelihood function is

$$p(y \mid \lambda) = \frac{\lambda^y e^{-\lambda}}{y!}.$$

Substituting the observed value $y = 7$:

$$p(y \mid \lambda) = \frac{\lambda^7 e^{-\lambda}}{7!}.$$

✓ ②

Since $7!$ does not depend on λ , we write

$$p(y \mid \lambda) \propto \lambda^7 e^{-\lambda}.$$

Step 3: Write down the prior density

The prior distribution is

$$\lambda \sim \Gamma(\alpha, \beta), \quad \alpha = 4, \beta = 2.$$

The Gamma density (rate parameterization) is

$$p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}.$$

Substituting $\alpha = 4, \beta = 2$:

$$p(\lambda) = \frac{2^4}{\Gamma(4)} \lambda^{4-1} e^{-2\lambda} = \frac{16}{\Gamma(4)} \lambda^3 e^{-2\lambda}.$$

Ignoring constants independent of λ , we write

$$p(\lambda) \propto \lambda^3 e^{-2\lambda}.$$

Step 4: Multiply likelihood and prior

Using Bayes' theorem:

$$p(\lambda | y) \propto p(y | \lambda) p(\lambda).$$

Substituting the expressions obtained above:

$$p(\lambda | y) \propto (\lambda^7 e^{-\lambda})(\lambda^3 e^{-2\lambda}).$$

Step 5: Combine like terms

Combine powers of λ :

$$\lambda^7 \cdot \lambda^3 = \lambda^{10}.$$

Combine exponential terms:

$$e^{-\lambda} \cdot e^{-2\lambda} = e^{-3\lambda}.$$

Hence,

$$p(\lambda | y) \propto \lambda^{10} e^{-3\lambda}.$$

Step 6: Identify the posterior distribution

The kernel of a Gamma distribution is

$$\lambda^{\alpha^*-1} e^{-\beta^* \lambda}.$$

Matching terms:

$$\alpha^* - 1 = 10 \Rightarrow \alpha^* = 11, \quad \beta^* = 3.$$

$$\lambda | y \sim \Gamma(11, 3)$$

b. Find the posterior mean $E(\lambda | y)$.

Solution

(b) Posterior mean

We compute the posterior mean of λ under the posterior distribution

$$\lambda | y \sim \Gamma(11, 3).$$

Step 1: Recall the definition of posterior expectation

By definition,

$$E(\lambda | y) = \int_0^\infty \lambda p(\lambda | y) d\lambda.$$

Step 2: Write the posterior density explicitly

For a Gamma distribution with shape parameter α^* and rate parameter β^* ,

$$p(\lambda | y) = \frac{(\beta^*)^{\alpha^*}}{\Gamma(\alpha^*)} \lambda^{\alpha^*-1} e^{-\beta^* \lambda}, \quad \lambda > 0.$$

Substituting $\alpha^* = 11$ and $\beta^* = 3$:

$$p(\lambda | y) = \frac{3^{11}}{\Gamma(11)} \lambda^{10} e^{-3\lambda}.$$

Step 3: Substitute into the expectation integral

$$E(\lambda | y) = \int_0^\infty \lambda \left(\frac{3^{11}}{\Gamma(11)} \lambda^{10} e^{-3\lambda} \right) d\lambda.$$

Factor out constants:

$$E(\lambda | y) = \frac{3^{11}}{\Gamma(11)} \int_0^\infty \lambda^{11} e^{-3\lambda} d\lambda.$$

Step 4: Evaluate the integral using the Gamma function

Recall the Gamma function identity:

$$\int_0^\infty x^{k-1} e^{-\beta x} dx = \frac{\Gamma(k)}{\beta^k}.$$

Here, $k = 12$ and $\beta = 3$, so

$$\int_0^\infty \lambda^{11} e^{-3\lambda} d\lambda = \frac{\Gamma(12)}{3^{12}}.$$

Step 5: Simplify the expression

Substitute into the expectation:

$$E(\lambda | y) = \frac{3^{11}}{\Gamma(11)} \cdot \frac{\Gamma(12)}{3^{12}} = \frac{\Gamma(12)}{3\Gamma(11)}.$$

Using the identity $\Gamma(n+1) = n\Gamma(n)$:

$$\Gamma(12) = 11\Gamma(11).$$

Thus,

$$E(\lambda | y) = \frac{11\Gamma(11)}{3\Gamma(11)} = \frac{11}{3}.$$

$$\boxed{E(\lambda | y) = \frac{11}{3}}$$

3. Suppose a single observation $x = 42$ is recorded from the model:

$$x \mid \theta \sim \mathcal{N}(\theta, \sigma^2), \quad \sigma^2 = 9.$$

Assume the prior distribution:

$$\theta \sim \text{Uniform}(0, 100).$$

Required:

- a. Write Bayes' theorem for $p(\theta \mid x)$. (2 mks)

Solution

(a) Bayes' theorem

We derive the posterior distribution of θ given the observation $x = 42$ using Bayes' theorem.

Step 1: State Bayes' theorem

Bayes' theorem expresses the posterior density as

$$p(\theta \mid x) = \frac{p(x \mid \theta) p(\theta)}{p(x)},$$

where

$$p(x) = \int_{-\infty}^{\infty} p(x \mid \theta) p(\theta) d\theta$$

is the marginal likelihood (normalizing constant).

- b. Show that the posterior has a Normal kernel. (7 mks)

Solution

(b) Normal kernel

We show explicitly that the posterior density has the kernel of a Normal distribution.

Step 1: Write down the likelihood function

The sampling model is

$$x \mid \theta \sim \mathcal{N}(\theta, \sigma^2), \quad \sigma^2 = 9.$$

The Normal likelihood is

$$p(x \mid \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x - \theta)^2}{2\sigma^2}\right].$$

Substituting $x = 42$ and $\sigma^2 = 9$:

$$p(x \mid \theta) = \frac{1}{\sqrt{18\pi}} \exp\left[-\frac{(42 - \theta)^2}{18}\right].$$

Since the normalizing constant does not depend on θ , we write

$$p(x \mid \theta) \propto \exp\left[-\frac{(\theta - 42)^2}{18}\right].$$

Step 2: Write down the prior density

The prior distribution is

$$\theta \sim \text{Uniform}(0, 100).$$

The corresponding prior density is

$$p(\theta) = \begin{cases} \frac{1}{100}, & 0 \leq \theta \leq 100, \\ 0, & \text{otherwise.} \end{cases}$$

Step 3: Substitute into Bayes' theorem

Substituting the likelihood and prior into Bayes' theorem:

$$p(\theta | x) \propto \exp\left[-\frac{(\theta - 42)^2}{18}\right] \times \begin{cases} \frac{1}{100}, & 0 \leq \theta \leq 100, \\ 0, & \text{otherwise.} \end{cases}$$

Step 4: Simplify the expression

Since $\frac{1}{100}$ is constant with respect to θ , it can be absorbed into the proportionality constant. Therefore,

$$p(\theta | x) \propto \exp\left[-\frac{(\theta - 42)^2}{18}\right], \quad 0 \leq \theta \leq 100,$$

and

$$p(\theta | x) = 0 \quad \text{for } \theta \notin [0, 100].$$

Step 5: Expand the quadratic term

Expand the square in the exponent:

$$(\theta - 42)^2 = \theta^2 - 84\theta + 1764.$$

Substitute into the exponent:

$$-\frac{(\theta - 42)^2}{18} = -\frac{1}{18}\theta^2 + \frac{84}{18}\theta - \frac{1764}{18}.$$

Step 6: Remove constants independent of θ

The term $-1764/18$ does not depend on θ and can be absorbed into the proportionality constant. Hence,

$$p(\theta | x) \propto \exp\left[-\frac{1}{18}\theta^2 + \frac{84}{18}\theta\right].$$

Step 7: Complete the square

Factor out $-\frac{1}{18}$:

$$-\frac{1}{18}(\theta^2 - 84\theta).$$

Complete the square inside the parentheses:

$$\theta^2 - 84\theta = (\theta - 42)^2 - 1764.$$

Substitute back:

$$-\frac{1}{18}(\theta - 42)^2 + \frac{1764}{18}.$$

Again, the constant term $1764/18$ is absorbed into the proportionality constant.

Step 8: Identify the Normal kernel

Thus, the posterior density is proportional to

$$\exp\left[-\frac{1}{18}(\theta - 42)^2\right].$$

Comparing with the kernel of a Normal distribution,

$$\exp\left[-\frac{1}{2\sigma^2}(\theta - \mu)^2\right],$$

we identify

$$\mu = 42, \quad 2\sigma^2 = 18 \Rightarrow \sigma^2 = 9.$$

Conclusion

Therefore, the posterior density has the kernel of a Normal distribution:

$$\theta \sim \mathcal{N}(42, 9).$$

- c. State the posterior distribution form and explain why it is truncated. (4 marks)

Solution

(c) Truncated Normal posterior

We now state the exact form of the posterior distribution and explain why it is a truncated Normal distribution.

Step 1: Recall the posterior kernel

From part (b), we established that the posterior density satisfies

$$p(\theta | x) \propto \exp\left[-\frac{(\theta - 42)^2}{18}\right], \quad 0 \leq \theta \leq 100,$$

and

$$p(\theta | x) = 0 \quad \text{for } \theta \notin [0, 100].$$

Step 2: Identify the untruncated Normal distribution

The kernel

$$\exp\left[-\frac{(\theta - 42)^2}{18}\right]$$

is the kernel of a Normal distribution with mean 42 and variance 9, since

$$\frac{1}{2\sigma^2} = \frac{1}{18} \Rightarrow \sigma^2 = 9.$$

If θ were allowed to take any real value, the posterior would be

$$\theta | x \sim \mathcal{N}(42, 9).$$

Step 3: Incorporate the support of the prior

However, the prior distribution is

$$\theta \sim \text{Uniform}(0, 100),$$

which assigns zero probability density outside the interval $[0, 100]$.

Since the posterior density is proportional to the product

$$p(\theta | x) \propto p(x | \theta)p(\theta),$$

it follows that

$$p(\theta | x) = 0 \quad \text{for all } \theta \notin [0, 100].$$

Step 4: State the posterior distribution

By definition, a truncated Normal distribution is a Normal distribution whose support is restricted to a finite interval.

Thus, the posterior distribution is

$$\theta | x \sim \mathcal{N}(42, 9) \text{ truncated to } [0, 100].$$

Interpretation

The likelihood favors values of θ near 42, while the Uniform prior restricts admissible values to $[0, 100]$. The posterior therefore retains the Normal shape induced by the likelihood but is truncated to respect the prior support.